Task 1. Install Minikube

Minikube allows you to run a local Kubernetes cluster.

Download: [Minikube Installation Guide]

Install kubectl: [Kubernetes kubectl Installation]

For Windows PowerShell, if using Chocolatey:

```
None
choco install minikube
choco install kubernetes-cli
```

Start Docker Desktop first, then execute:

```
None
minikube start --driver=docker
```

Check cluster status:

```
None
minikube status
```

Check nodes:

```
None
kubectl get nodes
```

Task 2. Create a Kubernetes Pod

Create a file named:

`music-pod.yaml`

Configuration:

```
None
apiVersion: v1
kind: Pod

metadata:
  name: spotify-nginx

spec:
  containers:
    - name: nginx
      image: nginx:latest
      ports:
        - containerPort: 80
```

Apply the YAML:

```
None
kubectl apply -f music-pod.yaml
```

Task 3. Check and delete the Pod

Check pods:

```
None
kubectl get pods
```

Display detailed information:

```
None
kubectl describe pod spotify-nginx
```

Delete the pod:

```
None
kubectl delete pod spotify-nginx
```

Verify deletion:

```
None
kubectl get pods
```

Task 4. Create a NodePort Service

Important: First recreate the pod because you deleted it in Task 3.

```
None
kubectl apply -f music-pod.yaml
```

Create a file:

`insta-service.yaml`

```
None
apiVersion: v1
kind: Service

metadata:
  name: insta-service

spec:
```

```
  type: NodePort

  selector:
    app: spotify-nginx

  ports:
    - port: 80
      targetPort: 80
      nodePort: 30080
```

One small correction: the Pod must have a matching label. Add this under its metadata:

```
None
metadata:
  name: spotify-nginx
  labels:
    app: spotify-nginx
```

Apply the service:

```
None
kubectl apply -f insta-service.yaml
```

Check service:

```
None
kubectl get services
```

Access using Minikube:

```
None
minikube service insta-service
```

Alternatively:

```
None
minikube ip
```

Open:

```
None
http://MINIKUBE-IP:30080
```

Task 5. Three advantages of Kubernetes

1. Auto-scaling: Zomato can increase application replicas when food orders increase.
2. Self-healing: Kubernetes automatically restarts failed containers.
3. Load balancing: Flipkart can distribute customer requests across multiple application pods.